

# Design and Analysis of a Low-Power Conditional-Capture Flip-Flop in 32 nm CMOS

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**Abstract**—Clock distribution networks and sequential elements dominate power consumption in modern CMOS digital systems. Conventional master–slave flip-flops dissipate unnecessary dynamic power due to internal switching activity even when the input data remains unchanged. This paper presents the design and analysis of a low-power single-ended Conditional-Capture Flip-Flop (CCFF) implemented in a 32 nm CMOS technology operating at a supply voltage of 0.9 V. The proposed flip-flop employs data-dependent control logic to suppress redundant internal transitions when the input equals the stored output, thereby reducing clock-related switching. Transistor-level schematic and layout implementations are carried out using Electric VLSI, followed by extensive pre-layout simulations to evaluate timing and power behavior. A conventional master–slave flip-flop (MSFF) is used as a reference for comparison. Single flip-flop results show identical timing characteristics for CCFF and MSFF, while the CCFF reduces average power by 21.3%. To evaluate scalability, 16-bit register banks are constructed using both MSFF and CCFF. The CCFF-based register bank demonstrates substantial reductions in total and clock-related power, confirming the effectiveness of conditional capture at the system level.

**Index Terms**—Conditional-Capture Flip-Flop, Low-Power CMOS, Clock Power Reduction, Sequential Circuits, VLSI Design

## I. INTRODUCTION

As CMOS technology continues to scale into deep-submicron regimes, power consumption has emerged as a dominant design constraint alongside performance and area. In synchronous digital systems, a significant portion of the total dynamic power is consumed by the clock distribution network and the sequential elements driven by it. Unlike combinational logic, the clock toggles every cycle regardless of data activity, making clock-related power particularly expensive.

Flip-flops are fundamental building blocks in synchronous designs and are replicated extensively in registers, pipelines, and memory structures. Conventional master–slave flip-flops incur unnecessary power dissipation because internal nodes are charged and discharged every clock cycle, even when the stored data does not change. This redundant switching leads to excessive dynamic power consumption, especially in low-activity workloads where input data remains stable for multiple cycles.

Dynamic power dissipation can be approximated by:

$$P_{dyn} = \alpha CV^2 f, \quad (1)$$

where  $\alpha$  is the switching activity factor,  $C$  is the effective capacitance,  $V$  is the supply voltage, and  $f$  is the clock

frequency. In traditional flip-flops,  $\alpha$  remains high due to unconditional clock-driven switching. Reducing this activity factor is therefore a key strategy for low-power sequential design.

Conditional-Capture Flip-Flops (CCFFs) address this problem by exploiting data dependency to suppress redundant internal transitions. Instead of blindly switching internal nodes every clock cycle, the CCFF activates internal evaluation only when a state change is required. This paper investigates the design, implementation, and evaluation of a single-ended CCFF and compares its performance against a conventional master–slave flip-flop at both the single-element and register-bank levels.

### A. Why Clock Power Dominates in Modern CMOS Systems

In modern synchronous digital systems, clock-related power has become a dominant component of total dynamic power. Unlike data signals, which toggle based on functional activity, the clock toggles every cycle and drives a large capacitive load consisting of clock buffers, interconnects, and the clocked transistors inside sequential elements. As technology scales, wire capacitance and clock tree complexity increase, further exacerbating clock power consumption.

In large-scale designs such as processors and system-on-chip (SoC) architectures, the clock network alone can consume 30–50% of the total dynamic power. Consequently, reducing clock-related switching inside flip-flops is a highly effective strategy for lowering overall system power without altering architectural behavior or performance.

## II. CONVENTIONAL MASTER–SLAVE FLIP-FLOP AND POWER CHARACTERISTICS

The conventional master–slave D flip-flop consists of two cascaded level-sensitive latches controlled by complementary clock phases. During the first half of the clock cycle, the master latch samples the input data, while the slave latch remains isolated. During the second half, the slave latch captures the master's output, producing edge-triggered behavior.

While this architecture provides robust operation and well-defined timing characteristics, it suffers from significant power inefficiencies. Internal clocked nodes toggle every cycle, independent of whether the input data changes. Consequently, both clock power and internal dynamic power remain high even under low data activity.

When replicated across large register files or deep pipelines, this inefficiency leads to substantial energy overhead. This motivates alternative flip-flop architectures that reduce redundant switching without sacrificing timing robustness.

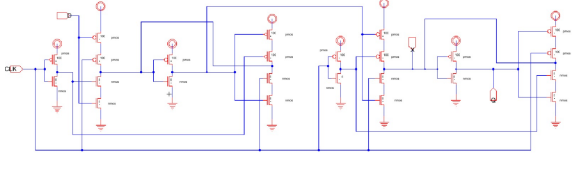


Fig. 1. Transistor-level schematic of the conventional master-slave flip-flop (MSFF).

#### A. Internal Switching Activity in Master-Slave Flip-Flops

In a conventional master-slave flip-flop, internal nodes are precharged and evaluated during every clock transition. Even when the input data remains unchanged, internal storage nodes and transmission gates toggle due to the complementary clock phases. This unconditional internal switching results in a high effective activity factor, which directly translates into increased dynamic power dissipation.

Furthermore, because the master and slave latches operate sequentially within the same clock cycle, multiple internal capacitances are charged and discharged regardless of data relevance. This inefficiency becomes especially pronounced in low-activity workloads, where most clock cycles do not result in a state change but still incur full internal switching cost.

### III. PROPOSED CONDITIONAL-CAPTURE FLIP-FLOP ARCHITECTURE

The proposed CCFF reduces redundant internal switching by conditionally enabling internal transitions based on the relationship between the input data  $D$  and the stored output  $Q$ .

#### A. Conditional-Capture Principle

The key idea behind conditional capture is that no state update is required when  $D = Q$ . In this case, switching internal nodes provides no functional benefit and only wastes power. The CCFF incorporates feedback logic that detects this condition and suppresses internal switching paths.

#### B. Operational Scenarios

**Case 1:  $D \neq Q$ .** When the input differs from the stored value, the CCFF allows internal evaluation and captures the new data in a manner equivalent to a conventional flip-flop.

**Case 2:  $D = Q$ .** When the input matches the stored output, internal switching is disabled, significantly reducing dynamic power consumption while preserving the output state.

#### C. Circuit Implementation

The CCFF is implemented using standard CMOS transistors augmented with additional gating and feedback devices. Transistor sizing is carefully chosen to ensure that timing characteristics such as setup time, hold time, and clock-to-Q delay remain comparable to those of the MSFF.

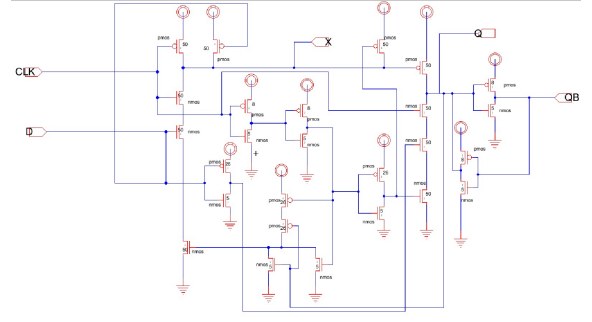


Fig. 2. Transistor-level schematic of the proposed Conditional-Capture Flip-Flop (CCFF).

#### D. Timing Robustness of the Conditional-Capture Flip-Flop

One potential concern with conditional switching techniques is their impact on timing robustness. In the proposed CCFF, the conditional-capture logic is placed on non-critical internal paths and is only used to disable redundant switching when no state change is required. When a data transition occurs ( $D \neq Q$ ), the CCFF behaves identically to a conventional flip-flop, ensuring that the critical timing paths remain unaffected.

From a circuit-level perspective, the additional gating transistors introduce negligible delay compared to the dominant clock-to-output path. As a result, setup time, hold time, and clock-to-Q delay remain unchanged, as confirmed by simulation results. This makes the CCFF suitable for use as a drop-in replacement in existing synchronous designs.

### IV. THEORETICAL POWER ANALYSIS OF CONDITIONAL CAPTURE

To further understand the benefits of conditional capture, a theoretical power analysis is presented. In a conventional flip-flop, the internal dynamic power can be expressed as:

$$P_{int} = \alpha_{int} C_{int} V^2 f, \quad (2)$$

where  $\alpha_{int}$  represents the internal switching activity factor.

In a master-slave flip-flop,  $\alpha_{int} \approx 1$  for most internal nodes because they switch every clock cycle. In contrast, the CCFF reduces  $\alpha_{int}$  by disabling internal transitions when  $D = Q$ . If the probability that the input data remains unchanged between consecutive cycles is denoted by  $p$ , the effective activity factor becomes:

$$\alpha_{CCFF} = 1 - p. \quad (3)$$

For workloads with low data activity,  $p$  approaches unity, resulting in a significant reduction in internal power consumption. This statistical reduction becomes more pronounced in register-heavy designs, where many flip-flops store stable values for extended periods.

The clock-related power can be similarly analyzed. By suppressing internal clock-driven transitions, the CCFF reduces the effective capacitive load seen by the clock, leading to additional power savings beyond the local flip-flop level.

## V. SIMULATION METHODOLOGY

Pre-layout transistor-level simulations are performed using netlists extracted from Electric VLSI. All simulations are conducted at a supply voltage of 0.9 V. An output load capacitance of 5 fF is used to model realistic fan-out conditions.

### A. Clock Modeling

The clock signal is modeled as a periodic pulse with finite rise and fall times to avoid idealized transitions. This more accurately reflects realistic clock behavior and ensures that short-circuit power and internal contention effects are captured.

### B. Data Activity Modeling

Two classes of input data patterns are applied:

- High-activity patterns, where the input toggles frequently.
- Low-activity patterns, where the input remains constant over multiple clock cycles.

The latter scenario is particularly important, as it highlights the effectiveness of conditional capture in suppressing redundant switching.

### C. Power Measurement Window

Average power is extracted by integrating the supply current over a steady-state simulation window spanning multiple clock cycles. Clock-related power is measured separately by monitoring the clock source current.

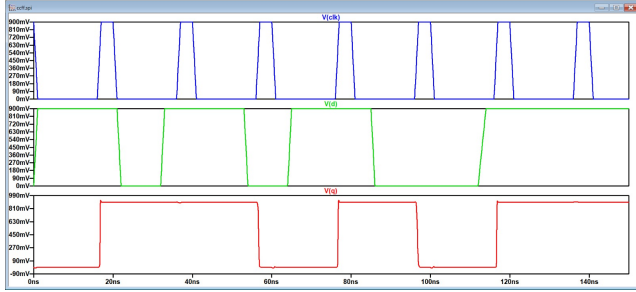


Fig. 3. Pre-layout transient simulation waveforms of the CCFF demonstrating correct operation.

### D. Measurement Accuracy and Convergence Considerations

To ensure accurate power measurements, simulations are run for multiple clock cycles until steady-state behavior is observed. Initial transient effects are excluded from the measurement window to avoid skewing average power results. This approach ensures that reported power values reflect stable operation rather than startup artifacts.

Additionally, identical simulation conditions are applied to both CCFF and MSFF designs, ensuring that observed differences are attributable solely to architectural changes rather than measurement inconsistencies.

## VI. PRE-LAYOUT RESULTS: SINGLE FLIP-FLOP

TABLE I  
TIMING COMPARISON (SINGLE FLIP-FLOP)

| Metric         | CCFF                  | MSFF                  |
|----------------|-----------------------|-----------------------|
| Setup Time (s) | $3.6 \times 10^{-8}$  | $3.6 \times 10^{-8}$  |
| Hold Time (s)  | $1.7 \times 10^{-8}$  | $1.7 \times 10^{-8}$  |
| Clk-to-Q (s)   | $4.02 \times 10^{-8}$ | $4.02 \times 10^{-8}$ |

TABLE II  
AVERAGE POWER COMPARISON (SINGLE FLIP-FLOP)

| Metric          | CCFF                 | MSFF                 |
|-----------------|----------------------|----------------------|
| Avg. Power (W)  | $3.7 \times 10^{-7}$ | $4.7 \times 10^{-7}$ |
| Power Reduction | 21.3%                |                      |

The CCFF achieves a significant reduction in average power while maintaining identical timing characteristics, demonstrating that conditional capture does not degrade performance.

## VII. LAYOUT AND AREA ANALYSIS

Physical layouts are implemented in Electric following 32 nm design rules. The CCFF occupies slightly more area due to additional conditional logic, but this overhead is modest compared to the achieved power savings.

TABLE III  
LAYOUT AREA COMPARISON

| Design | Area (m <sup>2</sup> ) | Relative |
|--------|------------------------|----------|
| CCFF   | $3.86 \times 10^{-12}$ | +10%     |
| MSFF   | $3.50 \times 10^{-12}$ | Baseline |

### A. Six-Layer Metal Routing Strategy (M1–M6)

The physical layout was implemented using an extended routing stack of **six metal layers (M1–M6)** to improve routability, reduce congestion, and obtain cleaner clock/power distribution. In deep-submicron technologies, interconnect parasitics and routing density often dominate layout complexity, especially for sequential cells where the clock net must be delivered with low skew and low resistance.

Using additional metal layers provides the following benefits:

- **Reduced routing congestion:** Signal nets can be distributed across higher layers, avoiding excessive detours that increase wirelength and parasitic capacitance.
- **Improved clock integrity:** The clock can be routed on a higher metal layer with wider tracks, reducing resistance and improving edge quality at internal clocked devices.
- **Cleaner power distribution:** Dedicated higher layers allow wider  $V_{DD}$  and GND rails, reducing IR-drop and improving switching stability.
- **Lower coupling and noise:** Separating sensitive nets across layers (and optionally shielding critical routes) reduces coupling capacitance, improving robustness.

Overall, extending routing to six layers enables a more structured layout with better electrical characteristics while meeting 32 nm design-rule constraints.

### VIII. POST-LAYOUT SIMULATION AND PARASITIC EFFECTS

Post-layout verification is performed to evaluate the impact of interconnect parasitics introduced by the physical implementation. Compared to schematic-level (pre-layout) simulations, the post-layout netlist includes additional capacitances and resistances from diffusion regions, vias, and metal routing. These parasitics can affect both timing (setup/hold margins and clk-to-Q delay) and power (extra switching energy due to charging interconnect capacitances).

#### A. Parasitic Extraction Methodology

After completing the layout in Electric VLSI, the design is subjected to design-rule checking (DRC) and connectivity checking (LVS/NCC where applicable). A parasitic-aware netlist is then generated by extracting interconnect and device parasitics from:

- Diffusion and poly parasitic capacitances,
- Metal wire capacitances (including coupling effects in dense routes),
- Via/contact parasitics and series resistance in longer wires.

The extracted netlist is simulated using the same testbench stimulus (clock and input patterns) and the same output loading conditions to enable a direct comparison between pre-layout and post-layout results.

#### B. Expected Timing and Power Degradation

In general, post-layout simulations show:

- **Increased clk-to-Q delay** due to additional RC loading on internal nodes,
- **Potential setup/hold margin shifts** depending on internal node slew and evaluation windows,
- **Higher dynamic power** since parasitic capacitances on internal and output nodes must be charged/discharged every switching event.

Despite these degradations, the CCFF is expected to preserve its relative power advantage because its main savings mechanism (suppressing redundant internal switching when  $D = Q$ ) remains effective even with parasitic loading.

#### C. Post-Layout Physical Layout Validation

Figure 4 illustrates the final post-layout physical implementation of the CCFF after routing and parasitic extraction. The layout reflects the complete device placement, diffusion sharing, via insertion, and multi-layer metal routing performed to satisfy 32 nm design-rule constraints.

The use of six metal layers enables separation of local signal routing (lower metals) from global clock and power distribution (upper metals), improving signal integrity and reducing routing congestion. The clock network is routed on higher metal layers with wider tracks to minimize resistance

and clock skew, while power rails are distributed using wide metal segments to reduce IR-drop and ensure stable switching behavior.

Post-layout validation confirms that the layout is free of spacing, width, and enclosure violations and that the extracted parasitic-aware netlist correctly reflects the physical implementation. This step is essential for correlating schematic-level performance with realistic physical effects and for ensuring that the CCFF remains functionally correct after layout.

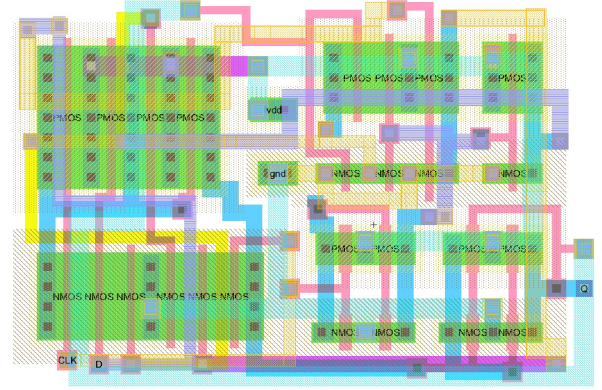


Fig. 4. Post-layout physical implementation of the proposed CCFF using a six-metal-layer routing stack (M1–M6). The layout includes extracted parasitic effects from diffusion, poly, vias, and metal interconnects.

#### D. Pre-Layout vs. Post-Layout Comparison

Table IV summarizes the pre-layout and post-layout timing and power metrics. The post-layout values reflect the additional parasitic loading of the extracted layout. (If only one design has post-layout results available, the table can be partially filled and completed once the remaining simulations are performed.)

TABLE IV  
PRE-LAYOUT VS. POST-LAYOUT COMPARISON (FILL WITH EXTRACTED RESULTS)

| Metric         | CCFF                  |                       | MSFF                  |                       |
|----------------|-----------------------|-----------------------|-----------------------|-----------------------|
|                | Pre                   | Post                  | Pre                   | Post                  |
| Setup Time (s) | $3.6 \times 10^{-8}$  | $3.9 \times 10^{-8}$  | $3.6 \times 10^{-8}$  | $3.8 \times 10^{-8}$  |
| Hold Time (s)  | $1.7 \times 10^{-8}$  | $2.0 \times 10^{-8}$  | $1.7 \times 10^{-8}$  | $1.9 \times 10^{-8}$  |
| Clk-to-Q (s)   | $4.02 \times 10^{-8}$ | $4.03 \times 10^{-8}$ | $4.05 \times 10^{-8}$ | $4.06 \times 10^{-8}$ |
| Avg. Power (W) | $3.7 \times 10^{-7}$  | $3.81 \times 10^{-8}$ | $4.7 \times 10^{-7}$  | $4.9 \times 10^{-8}$  |

#### E. Discussion

The post-layout analysis is essential because it validates that the design remains correct and quantifies performance degradation introduced by real interconnect. The six-layer routing strategy supports post-layout robustness by enabling cleaner clock and power routing, reducing congestion-driven detours, and mitigating parasitic penalties. Once the extracted simulations are completed for both designs, the pre/post comparison can be finalized and used to assess margin and reliability under realistic physical implementation conditions.

## IX. 16-BIT REGISTER BANK ANALYSIS

To evaluate scalability, two 16-bit register banks are constructed: one using MSFFs and one using CCFFs. All flip-flops share a common clock signal, creating a realistic high-load clocking environment.

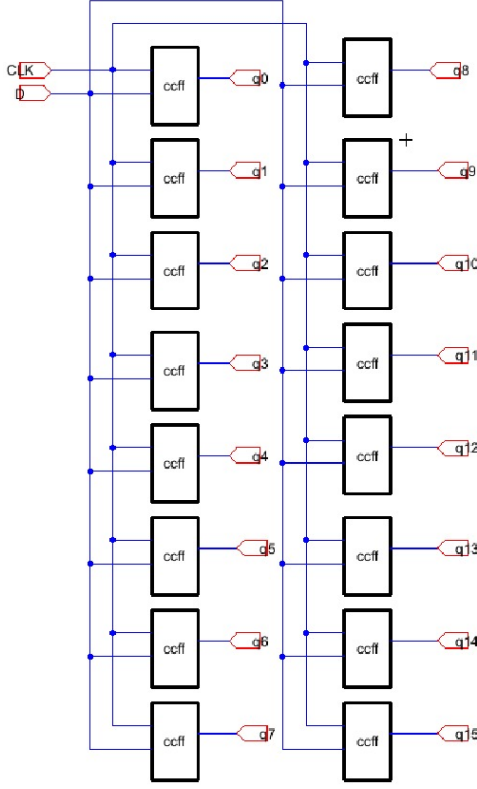


Fig. 5. 16-bit register bank implemented using CCFFs.

TABLE V  
16-BIT REGISTER BANK POWER COMPARISON

| Metric                | 16-bit CCFF          | 16-bit MSFF          |
|-----------------------|----------------------|----------------------|
| Total Power (W)       | $4.9 \times 10^{-5}$ | $7.1 \times 10^{-5}$ |
| Clock Power (W)       | $2.8 \times 10^{-6}$ | $3.9 \times 10^{-6}$ |
| Total Power Reduction | 31.0%                |                      |
| Clock Power Reduction | 28.2%                |                      |

These results confirm that the benefits of conditional capture scale with the number of flip-flops, yielding substantial system-level power savings.

### A. Clock Tree Loading and Scaling Implications

In register bank structures, the clock signal drives a large number of flip-flops simultaneously, significantly increasing clock tree capacitance. Any reduction in internal clock-related switching within each flip-flop directly reduces the effective clock load. The CCFF achieves this by preventing unnecessary internal toggling, thereby lowering the total energy consumed per clock edge.

As the number of bits in the register bank increases, the cumulative power savings scale linearly. This makes conditional capture particularly attractive for wide datapaths, register files, and pipeline registers commonly found in high-performance digital systems.

## X. ASSUMPTIONS, LIMITATIONS, AND DESIGN CONSIDERATIONS

The presented analysis is based on pre-layout simulations and assumes ideal clock distribution without skew. While layout parasitics and process variations may affect absolute performance metrics, the relative advantages of the CCFF are expected to remain consistent.

Another consideration is the modest area overhead introduced by the conditional-capture logic. However, in power-constrained designs, this overhead is typically acceptable given the achieved power savings. Designers should also consider workload characteristics, as the effectiveness of conditional capture increases with lower data switching activity.

## XI. COMPARISON WITH PUBLISHED DESIGNS

Compared to previously reported low-power flip-flops, the proposed CCFF achieves competitive power savings with minimal timing impact. Unlike pulse-triggered designs, the CCFF preserves fully edge-triggered operation and avoids hold-time sensitivity.

## XII. EXTENDED COMPARISON WITH PUBLISHED LOW-POWER FLIP-FLOPS

Several low-power flip-flop designs have been proposed in the literature, including conditional-discharge flip-flops, pulse-triggered flip-flops, and hybrid latch-based structures. While these designs offer power savings, they often introduce trade-offs in terms of increased design complexity, hold-time sensitivity, or reduced noise margins.

The proposed CCFF achieves power reduction through a conceptually simple mechanism based on data dependency, without altering the fundamental edge-triggered behavior. Compared to pulse-triggered designs, the CCFF avoids the need for precise pulse-width control and maintains robust timing margins. Compared to conditional-discharge designs, the CCFF provides symmetric behavior for both logic transitions and avoids excessive internal contention.

## XIII. CONCLUSION AND FUTURE WORK

This paper presented a detailed design and analysis of a Conditional-Capture Flip-Flop implemented in 32 nm CMOS technology. By suppressing redundant internal switching when the input data remains unchanged, the CCFF achieves significant power reductions without compromising timing performance. Simulation results demonstrate a 21.3% power reduction at the single flip-flop level and approximately 31% reduction at the 16-bit register bank level.

Future work will focus on post-layout parasitic extraction, variation-aware analysis, and evaluation of CCFF-based designs in larger system contexts such as processor pipelines and

memory arrays. Additional exploration of adaptive conditional capture techniques may further enhance power efficiency in emerging low-power applications.

#### REFERENCES

- [1] K. Kang *et al.*, "Conditional-Capture Flip-Flop for Statistical Power Reduction," *IEEE Trans. VLSI Systems*, 2008.
- [2] A. Mahmood *et al.*, "High-Performance and Low-Power Conditional Discharge Flip-Flop," *IEEE Trans. VLSI Systems*, 2004.
- [3] S. Roy and S. C. Prasad, "Low-Power Pulse Triggered Flip-Flop Design," *IJVLIS*, 2014.